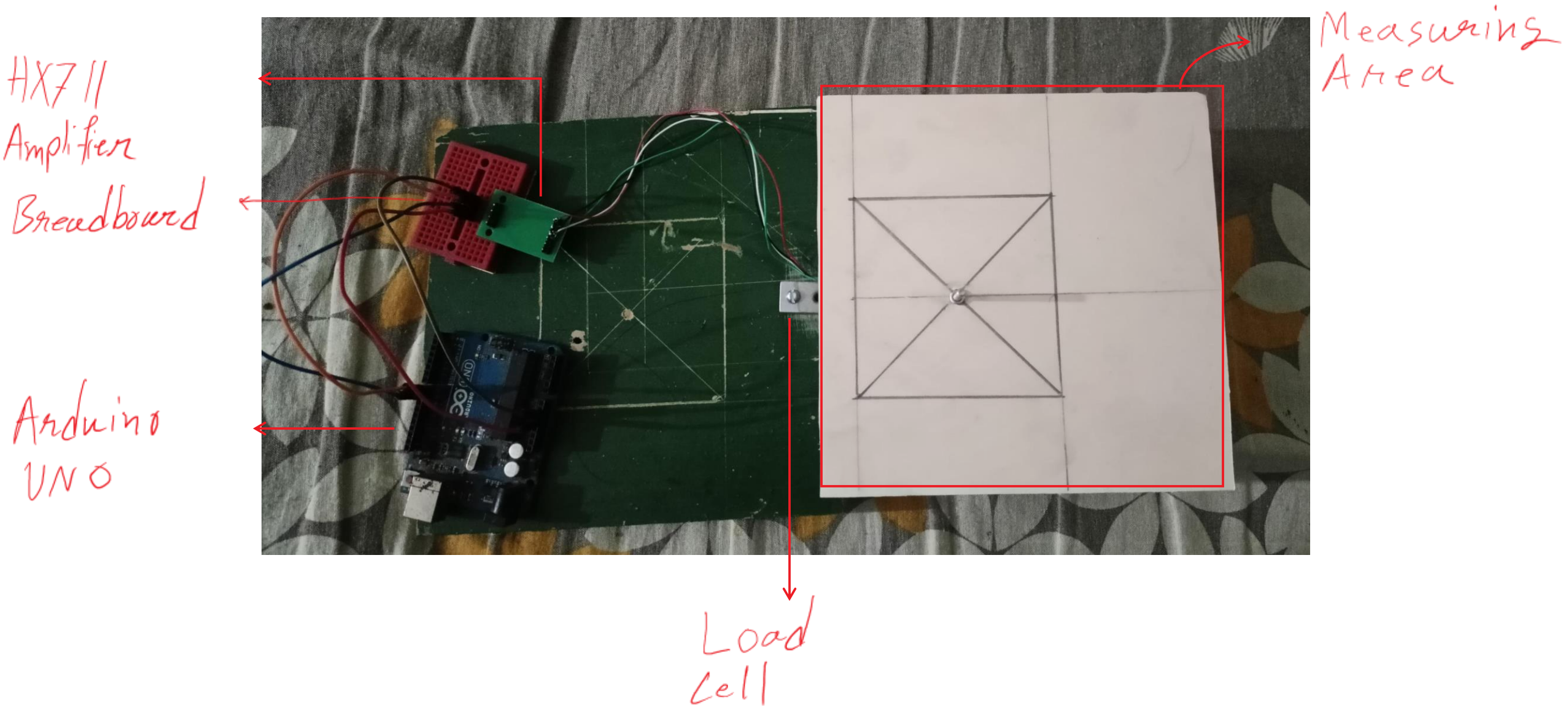
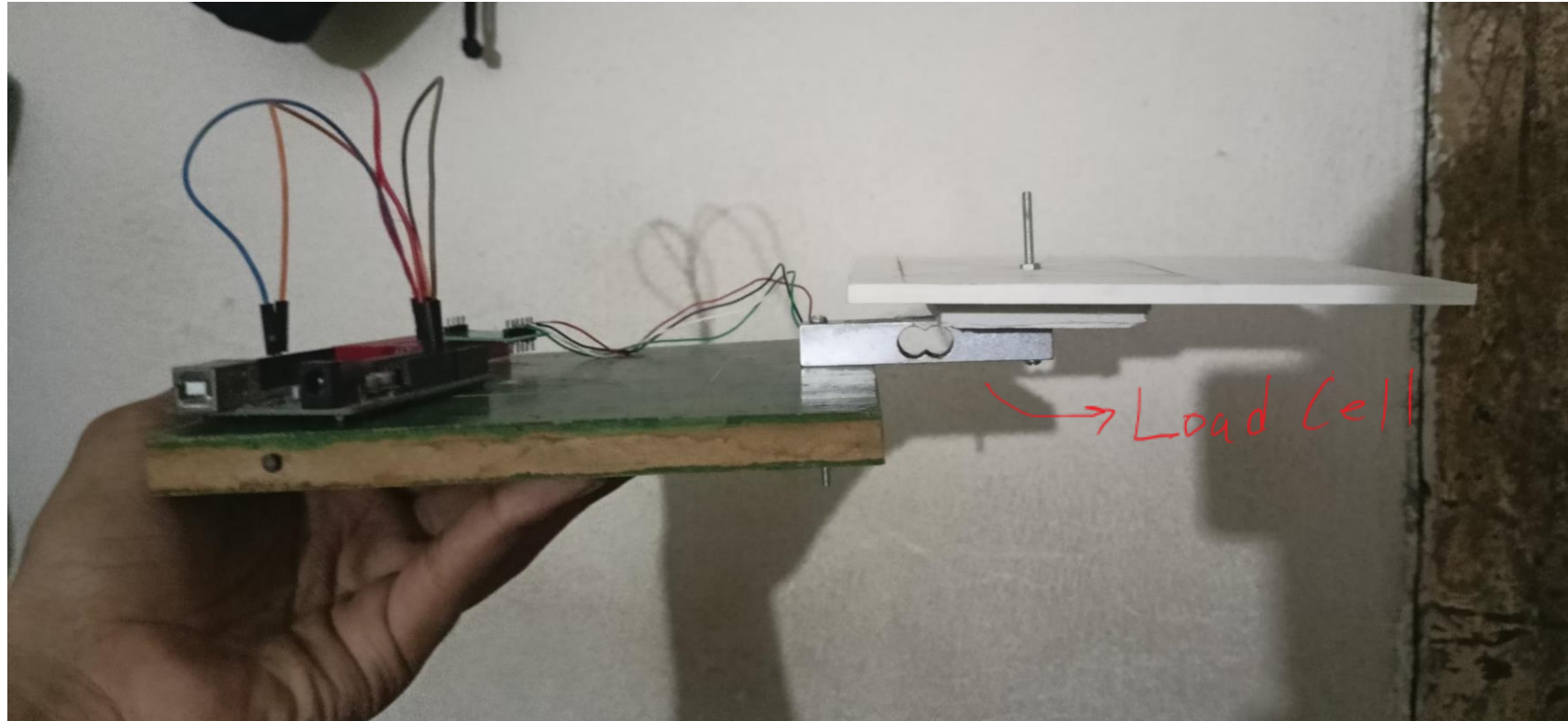


Test stand prototype (Top)

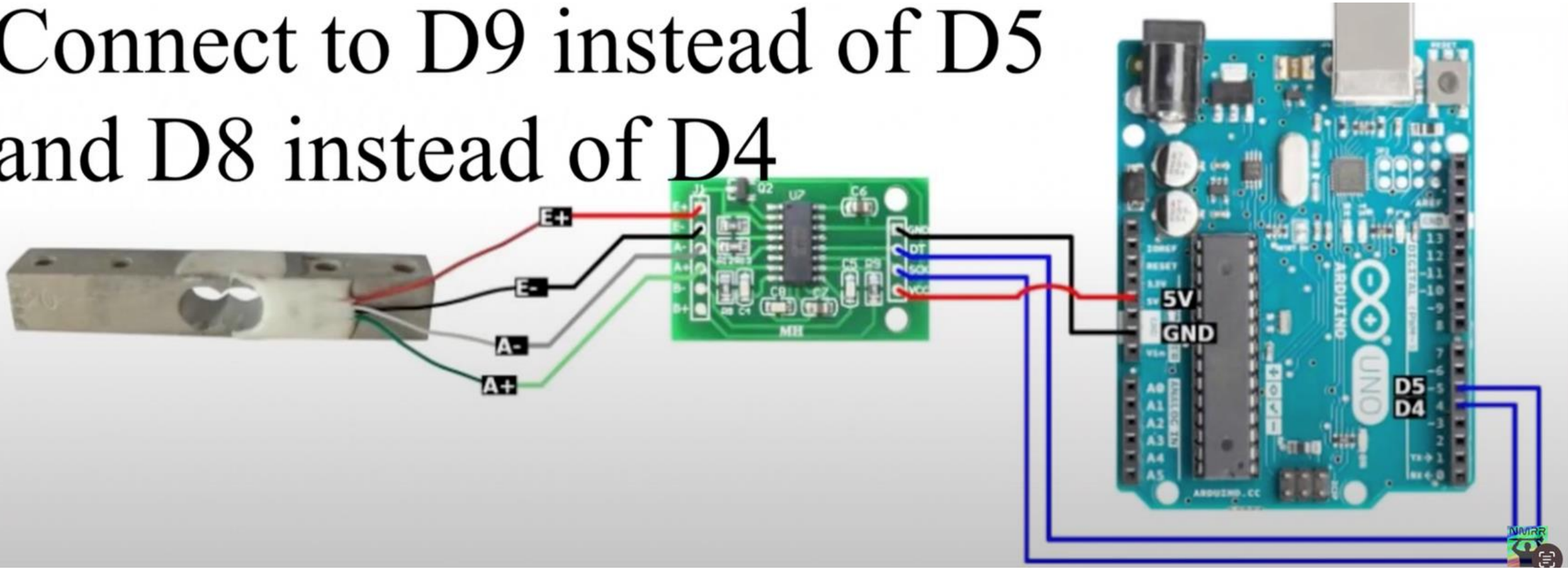


Side



Circuit diagram

Connect to D9 instead of D5 and D8 instead of D4



Working Procedure

Initially a menu is shown

For proper values calibration is required so we send 'r' to the arduino, which starts the calibration procedure.

Following the calibration procedure we get a calibration value

```
Model Rocket Test Stand
Model Rocket Test Stand Ready
Calibration factor: 220.00

===== MODEL ROCKET TEST STAND =====
Commands:
  t - Tare scale to zero
  r - Run calibration procedure
  c - Manually change calibration factor
  s - Start measurement/test
  x - Stop measurement/test
  p - Toggle serial plotter mode
  m - Show this menu
=====
```

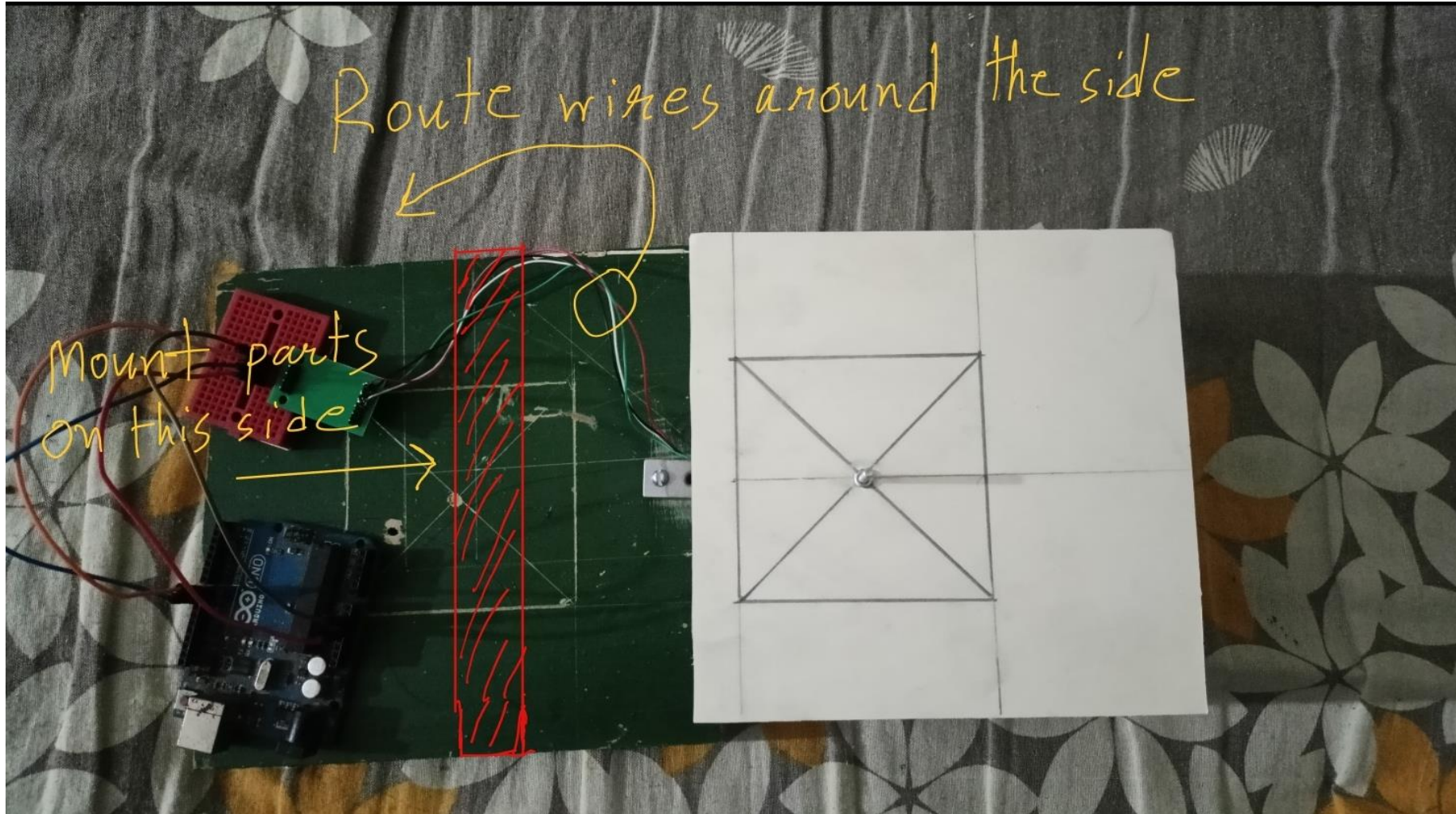
```
*** CALIBRATION PROCEDURE ***
Remove any load applied to the load cell.
Send 't' from the serial monitor to tare.
Tare complete
Place calibration weight
Send weight in grams into serial monitor
Known mass is: 185.00
New calibration value: 218.89
Save to EEPROM? (y/n): Value saved to EEPROM: 218.89
Calibration complete
```

Working Procedure

After calibration we remove the weight and tare the scale.
Then we start measurement and can see the current time,
weight and force of a sample

```
Taring...
Tare complete
MEASUREMENT STARTED
Time(s)  Weight(g)      Force(N)
378.601470    0.068527    0.000672
378.698791    0.077664    0.000762
378.796081    0.082232    0.000806
378.893402    0.091369    0.000896
378.990692    0.100506    0.000986
379.087982    0.095938    0.000941
379.185302    0.091369    0.000896
--- --
```


Required improvements



Required improvements

